

1 ORDER OF OPERATIONS

Enumerate: systeminfo, Get-ComputerInfo.
Users: Get-LocalUser. **Running:** Get-Process, Get-Service. **Ports:** netstat -ano. **Domain-joined?** Get-ComputerInfo | select CsPartOfDomain.

Firewall: Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True. Allow only what is required, block the rest.

User Management: Get-LocalUser, Get-LocalGroupMember Administrators. Disable unneeded accounts. Reset passwords on anything real.

Verify & Update: sfc /scannow. Windows Update GUI or Get-WindowsUpdate. DISM /Online /Cleanup-Image /RestoreHealth if sfc reports damage.

Secure Services: Get-Service | Where Status -eq Running. Disable unnecessary: Set-Service [name] -StartupType Disabled.

Enable Auditing: Advanced Audit Policy Configuration, or auditpol /set. Install Sysmon if allowed. Watch Event Viewer.

Deny red team tooling: Disable unused WinRM/PSRemoting, restrict admin shares, kill PowerShell v2 (see below).

2 USERS & GROUPS

net user [name] /add: create account.

- **net user [name] [pass] /add:** with password set.

- **net user [user] [newpass]:** change password.

Get-LocalUser — **Set-LocalUser:** PowerShell equivalent, scriptable.

- **Batch reset:** Get-LocalUser | Set-LocalUser -Password

(Read-Host -AsSecureString) in a loop.

net localgroup [group] [user] /add: add to group.

- **net user [user] /active:no:** disable.

- **net user [user] /active:yes:** reactivate.

whoami /priv: current user's privileges.

whoami /groups: group memberships, including hidden ones.

net accounts: view/set password and lockout policy locally.

runas /user:[user] cmd: run as another user.

Get-ChildItem -Path C:\Users -Recurse

-Filter authorizedkeys: find SSH keys if OpenSSH is installed.

query user / quser: logged in users and session IDs.

logoff [session ID]: force a session closed.

3 NETWORKING

ipconfig /all: interfaces and addresses.

netstat -ano: connections/routing, with owning PID.

- **Get-NetTCPConnection:** filterable, scriptable equivalent.

Windows Defender Firewall:

- **Status:** Get-NetFirewallProfile

- **Allow port:** New-NetFirewallRule -DisplayName X -Direction Inbound -LocalPort 443 -Protocol TCP -Action Allow

- **Block outbound by default:**

Set-NetFirewallProfile

-DefaultOutboundAction Block

- **List rules:** Get-NetFirewallRule | Where Enabled -eq True

- **Legacy CLI:** netsh advfirewall firewall add rule name="X" dir=in action=allow protocol=TCP localport=443

nmap [host(s)]: same external tool as **Linux**.

- **nmap -A [host]:** OS, service version, script scan.

- **Test-NetConnection [host] -Port [n]:** native port check, no nmap needed.

arp -a: local ARP cache, spot unexpected hosts.

Get-DnsClientCache: local DNS resolver cache.

4 FILE PERMISSIONS

icacls [file]: view or change permissions.

icacls [file] /grant [user]:(R,W): grant read/write.

icacls [file] /remove [user]: remove explicit permissions.

icacls [file] /inheritance:r: strip inherited permissions.

Get-Acl / Set-Acl: PowerShell equivalent, scriptable.

takeown /f [file]: take ownership.

attrib +h +s [file]: hidden/system flags; attackers hide with these too, worth checking for.

5 IMPORTANT FILES & LOCATIONS

...\config\SAM: local account database.

HKLM\HKCU ...\Run & RunOnce: common persistence.

...\winevt\Logs: Event Viewer logs on disk.

...\drivers\etc\hosts: local DNS overrides, check for tampering.

Startup folder: shell:startup, per-user and all-users.

HKLM\...\Winlogon\Shell: another classic persistence key.

Scheduled task XML: C:\Windows\System32\Tasks.

WMI event subscriptions: fileless persistence, check with Get-WmiObject -Namespace root\subscription -Class __EventFilter.

Get-WinEvent -LogName [log] -MaxEvents [n]: tail-like read of a log.

6 EVENT VIEWER & AUDITING

Setup: Local Security Policy → Advanced Audit Policy Configuration; enable logon, account management, process creation. Install Sysmon if allowed.

Key Event IDs:

- 4624 / 4625: successful / failed logon.
- 4634: logoff.
- 4720: user account created.
- 4732: member added to a local group.
- 4728: member added to a global group.
- 4688: new process (needs Audit Process Creation on).
- 5140: network share accessed.
- 1102: audit log cleared. Rare, always investigate.

Get-WinEvent -FilterHashtable

@{LogName='Security';Id=4625}: query by ID directly.

PowerShell script block logging: enables full command capture, huge for catching obfuscated PowerShell. Turn on via Group Policy → Windows Components → PowerShell.

7 SERVICES & SCHEDULED TASKS

Get-Service: list services and status.

- **Set-Service [name] -StartupType Disabled:** persistence/state.

- **Restart-Service [name]:** after a config change.

schtasks /query /fo LIST /v: all scheduled tasks, verbose.

Task Scheduler (taskschd.msc): check root and Microsoft → Windows folders; attackers often hide here.

Get-ScheduledTask — Where Author -notlike

"Microsoft*": filter to non-default tasks.

Get-CimInstance Win32_StartupCommand: another view of startup persistence.

8 MALWARE: DEFENDER & SYSINTERNALS

Windows Defender:

- **Get-MpComputerStatus:** confirm real-time protection on.

- **Update-MpSignature:** update signatures.

- **Start-MpScan -ScanType FullScan:** run a full scan.

- **Get-MpThreatDetection:** recent detections.

Sysinternals Suite (download ahead of time):

- **Autoruns / autorunsc.exe:** everything set to run at startup, CLI-friendly for scripting.

- **Process Explorer:** deeper than tasklist; parent/child, loaded DLLs.

- **TCPView:** GUI version of netstat.

- **Sigcheck:** verify a binary's digital signature before trusting it.

Yara on Windows: same rule syntax as **Linux**, works cross-platform.

9 PROCESSES & THREAT HUNTING

Get-Process: all running processes.

tasklist /svc: processes with hosted services.

taskkill /PID [pid] /F: kill a process.

taskkill /IM [name.exe] /F: kill all by image name.

Get-CimInstance Win32_Process -Filter

"ProcessId=[pid]": includes ParentProcessId, the Windows equivalent of tracing /proc/[pid].

- A shell (cmd.exe, powershell.exe) spawned from a service or web server process is not normal.

Sysmon Event ID 1: process creation with full command line and hash, if installed.

Get-Process — Select ProcessName,Path: flag processes running from unusual paths (Temp, user profile).

10 DENY RED TEAM: WINRM & LOLBINS

Confirm nothing legitimate depends on an item, then disable it and log the change.

- **WinRM/PSRemoting:** transport for Ansible's winrm connection and most PowerShell/Empire-style lateral movement. Disable if unused: Disable-PSRemoting -Force, block 5985/5986 at the host firewall.
- **Admin shares (C\$, ADMIN\$):** what psexec/wmiexec ride in on. Scope to your management host with a firewall rule instead of a blanket disable if your team relies on them.
- **PowerShell v2:** installable alongside v5+, lacks v5's logging; attackers downgrade to it on purpose. Disable-WindowsOptionalFeature -Online -FeatureName MicrosoftWindowsPowerShellV2.
- **LOLBins to watch/restrict via AppLocker:** certutil.exe -urlcache (download disguised as a cert op), regsvr32.exe /i:http... (Squiblydoo), mshta.exe, bitsadmin.exe /transfer. None normally reach the internet.
- **Compilers/scripting runtimes added for dev convenience (Python, Perl via Chocolatey):** remove if no scored service needs them; same Ansible/scripted-payload exposure as Linux.

11 ACTIVE DIRECTORY BASICS

Get-ADUser -Filter * -Properties *: every domain account.

Get-ADGroupMember "Domain Admins": who has full control.

dcdiag: domain controller health check.

repadmin /replsummary: replication status.

Get-GPO -All / gpresult /r: list GPOs, or confirm what applied.

nltest /domain_trusts: list domain trusts.

Common attack patterns:

- Kerberoasting: unusual TGS requests for service accounts (Event ID 4769).
- New accounts added directly to privileged groups.
- Unconstrained delegation with no real need.

12 RDP & REMOTE ACCESS

Enable/Disable RDP:

- Set-ItemProperty -Name fDenyTSConnections -Value 0 on the Terminal Server key: enable.
- Value 1 disables it.

Require Network Level Authentication: via Group Policy, or Set-ItemProperty on UserAuthentication.

Restrict who can connect: add only required users to Remote Desktop Users, not Administrators broadly.

Local Security Policy (secpol.msc): password policy, lockout policy, user rights assignment.

AppLocker / Software Restriction Policies: block execution from Temp or Downloads.

PSRemoting security: prefer HTTPS listener over the HTTP default; use constrained endpoints where possible.